1 Building End-to-End Supply Chain Visibility Using SQL

Introduction

FreshCart Pvt. Ltd. is a fast-growing consumer goods company. It operates several factories, regional warehouses, distributors and retail outlets across India. The company sells more than 800 SKUs, including packaged foods, beverages, home care and personal care products.

As the company expands, supply chain operations are becoming more complex. FreshCart generates data from purchase orders, supplier deliveries, warehouse scans, shipments and sales. However, this data is not being used properly. Several inefficiencies continue to affect daily operations.

The Challenge

FreshCart currently faces the following problems:

- Frequent shipment delays and inconsistent On Time In Full performance
- Stockouts in some regions even when total national inventory is enough
- Excess inventory in slow markets that increases holding cost
- Lack of clarity about which suppliers are reliable
- Limited insight into warehouse capacity, movement speed and bottlenecks
- No single place where supply chain KPIs are tracked or monitored

The company wants to use SQL-based analytics to create a clear view of its end-to-end supply chain.

SQL will help convert raw datasets into meaningful insights and help improve performance.

Your Mission

As project participants, your task is to design and implement a SQL-based supply chain visibility system.
The goal is to help FreshCart understand what is happening at every stage of the supply chain.
You are expected to:

- Create SQL queries to extract, clean and analyse supply chain data
- Find inefficiencies in supplier performance, warehouse movement and transportation
- Generate supply chain KPIs through SQL
- Provide insights that FreshCart can act on.

Submit both technical outputs and your analytical recommendations

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Key Tasks and Deliverables

SQL Queries

You will write and test SQL queries for:

- Fill Rate across warehouses and regions
- On Time In Full measurement for suppliers and transporters
- Order Cycle Time from purchase order creation to customer delivery
- Backorder identification
- Inventory Turnover and inventory ageing
- Warehouse inbound and outbound movement
- Delay patterns across routes and carriers

Database Optimization

You will also:

- Normalize the raw datasets into a clean relational structure
- Use SQL practices that improve performance such as indexing, joins, window functions and CTEs.
- Document your data cleaning and transformation decisions
- Create an ERD that shows how all the tables connect

Analytical Outputs

Using your SQL queries, you should prepare a report that:

- Identifies where the supply chain is slowing down
- Shows which SKUs move fast and which move slow
- Suggests inventory movement between regions
- Highlights unreliable suppliers or transport carriers
- Explains demand patterns

What We Expect From You

- Clean and well-structured SQL code
- Clear analytical thinking and problem solving
- Simple communication of your findings
- Effort to explore beyond the basic tasks if you see additional valuable insights

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Expected Business Impact

By completing this project, your SQL system should help FreshCart:

- Understand the full supply chain in one place
- Reduce stockouts
- Lower excess inventory and holding cost
- Improve supplier and logistics performance
- Make better, data-based decisions
- Improve the reliability of product availability for customers

Final Submission Should Include

- SQL scripts and documentation
- ERD or schema diagram
- A small KPI report or dashboard
- A short executive summary explaining your insights and recommendations

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Dataset

Dataset Link

Project Mentors

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Resource

Link

Whatsapp Group

Link